



Mangroves vs Waves



How can nature protect our coast from waves?

Waves can wash beaches away. Mangrove trees grow along the coast near Bangkok. Their roots stand in the water. Do they protect the land?

TIME

50–60 minutes

GROUPS

Groups of 3–4

AGES

Starter 6–8 · Explorer 9–11

Learning goals

- ✓ I can make a model beach and make waves the same way each time.
- ✓ I can see how roots change what waves do to sand.
- ✓ I can explain why mangroves are important.

STEAM in this activity

S

SCIENCE

Waves wash sand away: erosion.

T

TECHNOLOGY

A tray model shows a coast in miniature.

E

ENGINEERING

Compare ways to protect a beach.

A

ART

Draw a mangrove forest above and below the water.

M

MATHS

Count sticks and compare 3 with 15.



Before the lesson

- ☐ Deep trays with sides, damp sand, jugs of water and rulers.
- ☐ Lolly sticks or twigs: 15 per group.
- ☐ Cover the floor or work outside. Towels ready.
- ☐ Practise gentle waves so every group pushes the same way.



Safety

- Wipe up spills so nobody slips.
- Keep sand away from your eyes.
- Wash your hands after touching sand.



Kit per group

- | | | |
|--|---|--|
| ☐  A deep plastic tray | ☐  Damp sand | ☐  A jug of water |
| ☐  Towels for spills | ☐  10 sticks for mangrove roots | ☐  A ruler to make waves |
| ☐  15 sticks | | |

Running order (60 minutes)

0–10 min	Story: mangroves along the coast near Bangkok.
10–20 min	Build: sand beach and sea in the tray.
20–35 min	Test 1: no roots or roots.
35–50 min	Test 2: thin or thick forest.
50–60 min	Share and wash hands.

The investigations

Start here: build: Build your beach

Children record: Is your beach ready?; Draw your beach

What scientists often see: Where the land meets the sea is called the coast. Waves move sand along the coast every day.

Test 1: Do roots help?: Will "mangrove roots" stop the sand washing away?

Children record: How much sand washed away?; Draw the beach after the waves

What scientists often see: Mangrove roots slow the waves down and hold the mud and sand in place. Planting mangroves is one way to protect coasts from waves.

Test 2: Thin or thick forest?: Does a thicker mangrove forest protect the beach better?

Children record: How much sand washed away?; Draw the beach after the waves

What scientists often see: A wide, thick mangrove forest slows waves more than a thin line of trees. Scientists protect and replant mangroves to keep them thick.

Talk about it

- What happened to the sand when there were no roots?
- Why did we rebuild the beach the same way each time?
- What else could protect a beach? Is it better than trees?

What to look for

Starter: Makes waves and says whether more or less sand washed away.

Explorer: Explains that roots hold sand and slow waves, using their own results.

Art corner

Mangrove home. Draw a mangrove forest. Show the roots under the water, and the crabs, fish and birds that live there.

Maths corner

Starter: Draw 3 sticks and 15 sticks. Which is more?

Explorer: $15 - 3 = \underline{\quad}$ more sticks. Measure how far the waterline moved, in cm: $\underline{\quad}$.

Real-world link

In Thailand, people plant young mangrove trees along the coast to help protect the land from waves.

Going further

Build a small wall of stones instead of sticks. Which protects the beach better?

